

Problem Set 2 Write Up

Data and Methodology

Monthly value-weighted returns for 17 Fama–French industry portfolios were obtained from Ken French's Data Library (17_Industry_Portfolios.csv). Monthly returns for the five Fama–French (2015) factors — Market excess return (Mkt-RF), Size (SMB), Value (HML), Profitability (RMW), and Investment (CMA) — along with the monthly risk-free rate were drawn from the companion file (F-F_Research_Data_5_Factors_2x3.csv). Excess returns for each of the 17 industry portfolios were computed by subtracting the monthly risk-free rate from the corresponding gross return. The full sample used for summary statistics covers 1963:07 to 2026:02 (751 months). Summary statistics are annualized by multiplying the sample mean by 12 and the sample standard deviation by $\sqrt{12}$; the annualized Sharpe ratio is $\sqrt{12} \times \bar{r}_e / \hat{\sigma}$, where $\bar{r}_e$ is the monthly mean excess return and $\hat{\sigma}$ is the monthly sample standard deviation.

Table 1: Annualized Summary Statistics — 17 Industry Portfolio Excess Returns (1963:07–2026:02)

Industry	Mean (%)	Volatility (%)	Sharpe Ratio
Food	7.72	14.62	0.53
Mining & Minerals	9.13	25.78	0.35
Oil & Petroleum Products	8.17	21.10	0.39
Textiles, Apparel & Footwear	7.41	21.74	0.34
Consumer Durables	4.71	19.80	0.24
Chemicals	6.34	19.80	0.32
Drugs, Soap, Perfumes, Tobacco	8.64	15.38	0.56
Construction & Construction Materials	8.73	21.04	0.41
Steel Works Etc.	6.15	27.02	0.23
Fabricated Products	8.37	18.98	0.44
Machinery & Business Equipment	9.46	22.25	0.43

Automobiles	8.57	24.97	0.34
Transportation	8.05	19.83	0.41
Utilities	5.87	14.05	0.42
Retail Stores	8.56	18.25	0.47
Banks, Insur. Cos., & Other Financials	7.51	18.92	0.40
Other	6.86	16.57	0.41

Discussion

Across the 751-month sample, all 17 industry portfolios deliver positive annualized mean excess returns, ranging from 4.71% (Consumer Durables) to 9.46% (Machinery & Business Equipment), reflecting the broad equity risk premium. Machinery & Business Equipment and Mining & Minerals earn the highest mean excess returns at 9.46% and 9.13% respectively, consistent with their cyclical nature and higher exposure to economic growth. On the risk side, Steel Works Etc. is the most volatile industry at 27.02% annualized, followed by Mining & Minerals at 25.78% and Automobiles at 24.97% — all highly cyclical sectors sensitive to commodity prices and consumer spending cycles. By contrast, Food and Utilities exhibit the lowest volatility (14.62% and 14.05% respectively), reflecting their defensive, non-cyclical characteristics.

In terms of risk-adjusted performance, Drugs, Soap, Perfumes & Tobacco achieves the highest Sharpe ratio at 0.56, benefiting from relatively low volatility and a strong mean return. Food follows at 0.53. Consumer Durables and Steel Works Etc. produce the weakest Sharpe ratios (0.24 and 0.23 respectively), as their high volatility is not sufficiently compensated by their mean returns. The spread in Sharpe ratios across industries (0.23 to 0.56) is notable and suggests meaningful cross-sectional variation in risk-adjusted performance, which a multifactor model may be able to exploit.

Table 2: Annualized Summary Statistics — Fama-French Factor Returns (1963:07–2026:02)

Factor	Mean (%)	Volatility (%)	Sharpe Ratio
MKT (market)	7.11	15.43	0.46
SMB (size)	2.21	10.48	0.21
HML (value)	3.53	10.29	0.34
RMW (profitability)	3.20	7.69	0.42
CMA (investment)	2.99	7.17	0.42

Discussion

The market factor (Mkt-RF) dominates in terms of both mean return and volatility, earning 7.11% annualized with a Sharpe ratio of 0.46, consistent with the well-documented equity risk premium over the long run. The four non-market factors — SMB, HML, RMW, and CMA — all deliver positive mean returns, providing empirical support for the Fama-French (2015) five-factor model.

SMB (size premium) earns the lowest mean return at 2.21% with a Sharpe ratio of only 0.21, suggesting that the size effect has weakened considerably over this sample period, consistent with the broader academic literature. HML (value) earns 3.53% with a Sharpe ratio of 0.34, reflecting a moderate value premium. RMW (profitability) and CMA (investment) are particularly noteworthy: both earn Sharpe ratios of 0.42 — nearly as high as the market factor — despite having much lower volatility (7.69% and 7.17% respectively). This indicates that the profitability and investment factors offer attractive risk-adjusted returns with relatively low standalone volatility, making them valuable components of the multifactor forecasting framework used in portfolio construction.

Portfolio Construction

The total sample is divided into two periods:

- In-sample (IS): 1963:07 – 1974:12 (138 months), used to initialize the estimation windows
- Out-of-sample (OOS): 1975:01 – 2026:02 (613 months), used for real-time portfolio construction and evaluation

To simulate an investor operating in real time, for each month t in the OOS period, forecasts of the 17-dimensional vector of expected excess returns ($\hat{\mu}$) and the 17×17 covariance matrix ($\hat{\Sigma}$) are computed using only data available through month $t-1$. The forecasting procedure uses the Fama-French 5-factor model as follows:

Step 1 — Factor mean and covariance (expanding window): At each month t , the expected factor return vector $\hat{\mu}_f$ (5×1) is estimated as the sample mean of all factor returns from 1963:07 through $t-1$. The factor covariance matrix $\hat{\Sigma}_f$ (5×5) is estimated as the sample covariance matrix over the same expanding window. Using an expanding window ensures that the estimates incorporate all available historical information at each point in time.

Step 2 — Alpha, beta, and idiosyncratic variance (rolling 120-month window): For each of the 17 industry portfolios, a multifactor OLS regression is estimated using the 120 most recent monthly observations available at time t . This yields a 17-dimensional vector of intercepts $\hat{\alpha}$ and a 5×17 matrix of factor loadings $\hat{B}$. The OLS residual variances from each regression fill the diagonal of $\hat{\Sigma}_e$, the 17×17 diagonal idiosyncratic covariance matrix. The rolling window allows the factor exposures to adapt to changing market conditions over time.

Step 3 — Forecasts: The expected excess return vector and covariance matrix are constructed as:

$$\begin{aligned}\hat{\mu} &= \hat{\alpha} + \hat{B}'\hat{\mu}_f \\ \hat{\Sigma} &= \hat{B}'\hat{\Sigma}_f\hat{B} + \hat{\Sigma}_e\end{aligned}$$

The first term in $\hat{\Sigma}$ captures systematic (factor-driven) risk; the second captures idiosyncratic risk.

Step 4 — Portfolio optimization: At the end of each month $t-1$, the optimal weight vector w is determined by solving the expected utility maximization problem:

$$\max_w w'rf + w'\hat{\mu} - 0.5\gamma w'\hat{\Sigma}w \text{ subject to } 0 \leq w_i \leq 0.25 \text{ for } i = 1, \dots, 17$$

where $\gamma = 5$ is the coefficient of relative risk aversion and rf is the most recently observed risk-free rate. The constraint $0 \leq w_i \leq 0.25$ has two components: the lower bound ($w_i \geq 0$) prohibits short selling any industry portfolio, and the upper bound ($w_i \leq 0.25$) prevents any single industry from exceeding

25% of the total portfolio, enforcing diversification and limiting concentration risk. Note that the weights need not sum to one — any uninvested portion is implicitly held in the risk-free asset, which earns r_f and is already accounted for in the objective function.

Benchmark: In addition to the MV optimal portfolio, a simple equal-weighted (1/N) benchmark is constructed each month by assigning weight $1/17 \approx 5.88\%$ to each of the 17 industry portfolios. This benchmark requires no estimation and serves as a natural performance hurdle.

Table 3: Annualized Summary Statistics — Portfolio Excess Returns (1975:01–2026:02)

Portfolio	Mean (%)	Volatility (%)	Sharpe Ratio
Benchmark (1/N)	9.49	16.27	0.58
MV Optimal	6.76	12.71	0.53

Discussion of OOS Portfolio Performance

Over the 613-month OOS period from 1975:01 to 2026:02, both portfolios deliver meaningfully positive risk-adjusted performance, though the results reveal an interesting tradeoff between raw return and risk control. The 1/N benchmark earns a strong annualized mean excess return of 9.49% with a volatility of 16.27% and a Sharpe ratio of 0.58. The MV optimal portfolio earns a lower mean excess return of 6.76%, but achieves substantially reduced volatility of 12.71%, resulting in a Sharpe ratio of 0.53 — only modestly below the benchmark.

The MV portfolio's lower mean return relative to 1/N is consistent with the design of the optimization: the expected utility objective with $\gamma = 5$ penalizes variance heavily, so the optimizer tilts toward lower-risk allocations and selectively underweights industries with high estimated covariance, even at the cost of some mean return. The 25% cap on individual weights prevents extreme concentration, while the 0% floor prohibits short selling, together producing a well-diversified but more conservatively positioned portfolio than the fully equity-invested 1/N benchmark.

The fact that the MV portfolio achieves a Sharpe ratio of 0.53 versus 0.58 for 1/N — a gap of only 0.05 — is notable. It demonstrates that the multifactor regression framework, despite introducing estimation error through rolling OLS regressions, still produces a portfolio that competes meaningfully with naive diversification on a risk-adjusted basis. The MV portfolio's 3.56 percentage point reduction in annualized volatility (from 16.27% to 12.71%) represents a substantial improvement in risk control that would be highly valued by risk-averse investors.

Out-of-sample annualized CER ($\gamma = 5.0$):

- CER for MV Optimal: 2.7262%
- CER for Benchmark 1/N: 2.8778%
- Annualized CER Gain (MV vs. 1/N): -0.1516%

Certainty Equivalent Return (CER) Analysis

The annualized CER gain of -0.1516% indicates that the MV optimal portfolio falls just marginally short of the $1/N$ benchmark on a utility-adjusted basis for an investor with $\gamma = 5$. The CER for the MV portfolio (2.7262%) is only 0.15 percentage points below the $1/N$ benchmark (2.8778%) — an economically negligible difference. This near-zero CER gap tells a more nuanced story than the raw return comparison: while the $1/N$ benchmark earns a higher mean excess return, the MV portfolio's substantially lower variance nearly fully compensates for this shortfall from a utility perspective.

In practical terms, a risk-averse investor with $\gamma = 5$ would be nearly indifferent between the two strategies, needing only 0.15% per year in additional return to prefer the MV portfolio over $1/N$. This finding suggests that the Fama-French 5-factor multifactor regression framework provides genuine economic value in terms of risk reduction, even if it does not fully overcome the challenge of estimation error in expected return forecasts. The result is consistent with the broader literature: sophisticated forecasting models tend to add value primarily through better covariance estimation and risk control rather than through superior mean return forecasts, and their utility-based advantage over naive diversification is often modest but real.